

Minh-Nhat Nguyen

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PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate with a solid mathematical foundation (**Calculus, Linear Algebra, Stochastic Systems**) and hands-on research in **Computer Vision** and **Generative Models (Diffusion Models, Flow Matching)**. Experienced in developing training-free generative methods and high-throughput multimodal video retrieval pipelines. Seeking an **AI Engineer / Researcher** position in corporate **R&D** to advance applied computer vision and next-generation generative AI systems.

EDUCATION

University of Science, VNUHCM

Ho Chi Minh City, Vietnam

Bachelor of Information Technology – GPA: 3.74/4.0 (8.75/10.0)

Aug. 2025 – 2029 (expected)

Intended Major: Computer Science

Relevant Coursework (All 4.0/4.0): Calculus I & II, Linear Algebra, Data Structures & Algorithms, Programming Fundamentals

Advanced Online Coursework: MIT 6.S184: Generative AI with Stochastic Differential Equations

Le Quy Don Gifted High School

Da Nang, Vietnam

High School Diploma – Mathematics Specialization

Aug. 2022 – May 2025

EXPERIENCE & RESEARCH

Undergraduate AI Researcher

March 2026 – Present

University of Science, VNUHCM

Ho Chi Minh City, Vietnam

- Developing a Training-Free method for Style Transfer using Latent Diffusion Models and Flow Matching, introducing latent patch-shuffling to eliminate content leakage and pixel-space grid artifacts.
- Optimizing generative algorithms by manipulating sampling trajectories, guidance terms, and cross-attention maps to control stylistic and structural outputs.

HONORS & AWARDS

Quarter-finalist (Top 36 / 200+)

May 2026

Coding Challenge 2026 – Faculty of Information Technology, HCMUS

Ho Chi Minh City, Vietnam

PROJECTS

Video Retrieval Pipeline – AI Challenge HCMC 2026 | *LanceDB, SigLIP, DP*

Aug. 2026 – Present

- Leading a team in the Ho Chi Minh City AI Challenge 2026 to build an end-to-end multimodal video retrieval pipeline addressing Known-Item Search (KIS), Visual Question Answering (VQA), and Temporal Alignment (TRAKE).
- Engineered a high-throughput hybrid retrieval engine combining dense semantic embeddings (CLIP/SigLIP) with full-text search (BM25 via LanceDB) and Dynamic Programming (DANTE algorithm) for temporal sequence matching.
- Optimized candidate frame ranking, automated query parsing, and sub-second retrieval latency on large-scale video keyframe datasets.

Latent Patch-Shuffle Style Transfer | *PyTorch, Stable Diffusion, Flow Matching* | [Article Link](#)

Aug. 2026

- Researched and implemented a training-free style transfer framework on latent diffusion models, utilizing latent patch-shuffling to retain core object geometry while preventing content leakage.
- Eliminated pixel-space grid line artifacts by shifting patch permutations into the latent manifold, achieving vivid artistic style adaptation across diverse domains (Cyberpunk, Starry Night, Poster).

Handwritten Digits Classification | *PyTorch, CNN, MLP* | [GitHub](#)

Nov. 2025

- Led a research team to build and train Convolutional Neural Network (CNN) and Multilayer Perceptron (MLP) architectures from scratch for the MNIST dataset.
- Emphasized rigorous evaluation by benchmarking and comparing the efficiency, loss convergence, and accuracy between the two architectures.

- Co-authored an article detailing mathematical derivations and comparative analysis ([Article Link](#)).

Gomoku (Tic-Tac-Toe) AI Bot | *C++*, *Algorithms* | [GitHub](#)

Dec. 2025

- Developed a classical AI opponent for the game of Gomoku using procedural programming in C++.
- Implemented the Minimax algorithm combined with Alpha-Beta pruning to optimize decision-making processes and evaluate game states efficiently.

TECHNICAL SKILLS

Core Knowledge: Deep Learning, Computer Vision, Generative Models (Flow Matching, Diffusion Models, SDEs), Multimodal Information Retrieval (LanceDB, FAISS, Vector Search, BM25)

Programming & Tools: Python, C/C++, PyTorch, LanceDB, HuggingFace (Transformers/CLIP), Git, LaTeX

Mathematics: Linear Algebra, Multivariable Calculus, Optimization, Stochastic Differential Equations (SDEs)